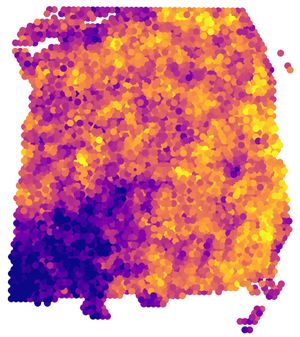
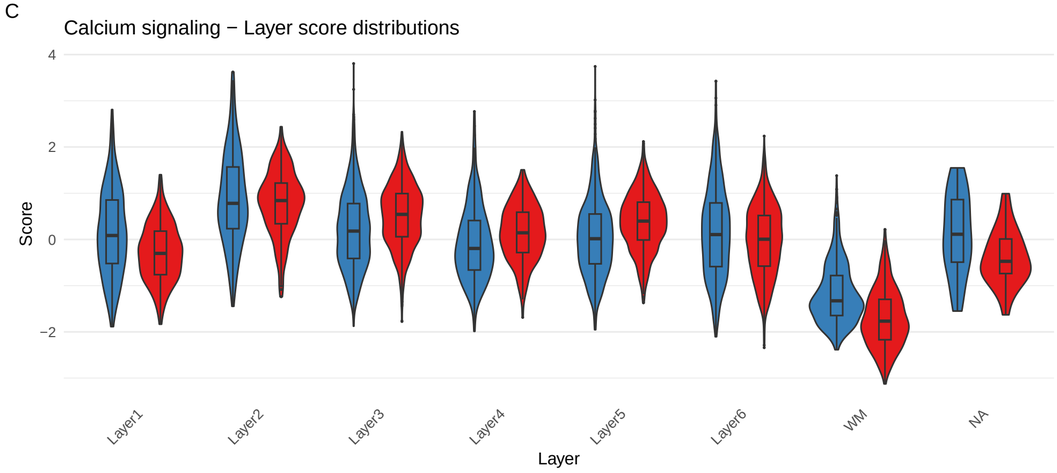
A


Calcium signaling−sGSVA

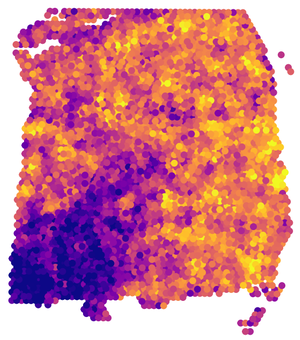
B


Calcium signaling−GSVA

C


Calcium signaling − Layer score distributions
Score
4
2
0
−2
Layer1
Layer2
Layer3
Layer4
Layer5
Layer6
WM
NA
Layer

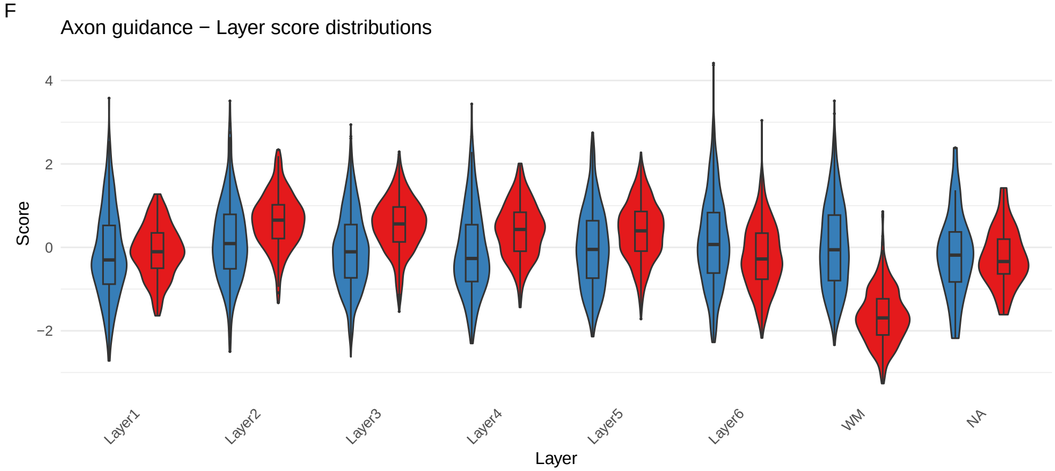
D


Axon guidance−sGSVA

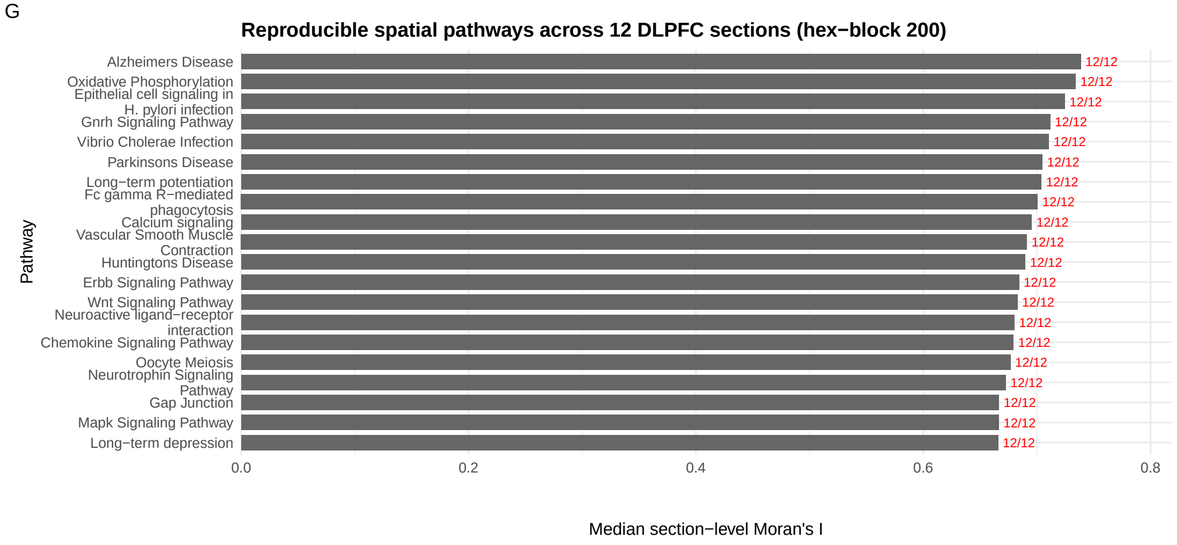
E


Axon guidance−GSVA

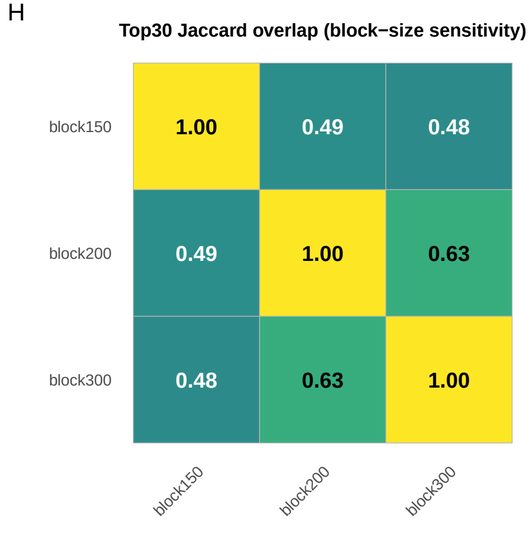
F


Axon guidance − Layer score distributions
Score
4
2
0
−2
Layer1
Layer2
Layer3
Layer4
Layer5
Layer6
WM
NA
Layer
sGSVA
1
0
−1
−2
GSVA
1
0
−1
−2
method
GSVA
sGSVA
Value
1.00
0.75
0.50
0.25
0.00

G


Reproducible spatial pathways across 12 DLPFC sections (hex−block 200)
Pathway
Alzheimers Disease
Oxidative Phosphorylation
Epithelial cell signaling in H. pylori infection
Gnrh Signaling Pathway
Vibrio Cholerae Infection
Parkinsons Disease
Long−term potentiation
Fc gamma R−mediated phagocytosis
Calcium signaling
Vascular Smooth Muscle Contraction
Huntingtons Disease
Erbb Signaling Pathway
Wnt Signaling Pathway
Neuroactive ligand−receptor interaction
Chemokine Signaling Pathway
Oocyte Meiosis
Neurotrophin Signaling Pathway
Gap Junction
Mapk Signaling Pathway
Long−term depression
12/12
0.0
0.2
0.4
0.6
0.8
Median section−level Moran's I

H


Top30 Jaccard overlap (block−size sensitivity)
block150
block200
block300
1.00
0.49
0.48
0.49
1.00
0.63
0.48
0.63
1.00

I


Rank correlation (block−size sensitivity)
block150
block200
block300
1.00
0.37
0.34
0.37
1.00
0.66
0.34
0.66
1.00